Unit 1

Campbell Biology, AP Edition, 12e

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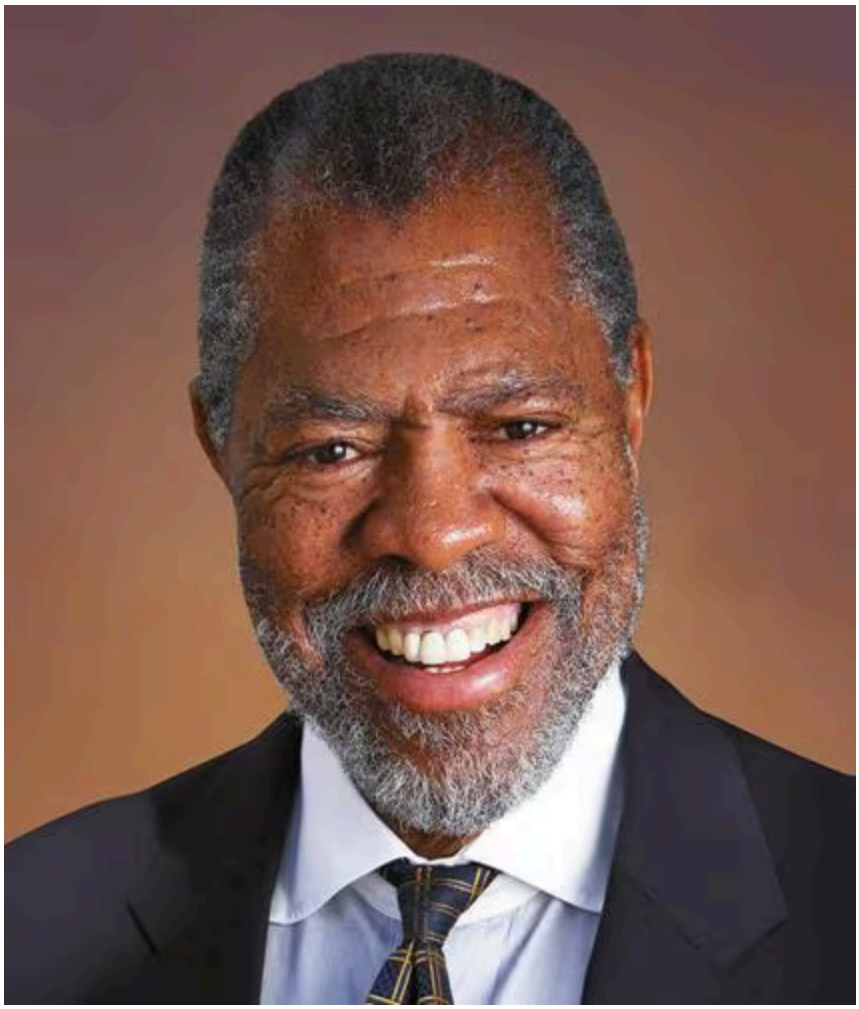
The Chemistry of Life

Dr. Kenneth Olden



Dr. Kenneth Olden recently retired from a long and distinguished career in medical research and public health, including serving as Director of the National Institute of Environmental Health Sciences from 1991 to 2005, as founding Dean of the School of Public Health at the City University of New York from 2008 to 2012, and as Director of the National Center for Environmental Assessment from 2012 to 2016. He has published over 220 research papers and has received many honors and awards, among them the three most distinguished awards in public health. Dr. Olden grew up in Parrottsville, Tennessee, the son of a

sharecropper. He remembers walking up a long hill to high school every morning and daydreaming about helping the poor people—both black and white—in the neighborhoods he'd walk through, wanting to make a difference.

An Interview with: Kenneth Olden

What got you interested in biology?

I was always cerebral, I always liked to read and think. For me, role models were important. At that time, I knew about only two professions that blacks were in: medicine and teaching. There was one black physician in town—unusual for a rural town. My high school principal—he was black—would tell us, "By golly, you could be anything you want to be!" I paid attention and listened. He helped me apply to Knoxville College, and I decided I would be a physician, so I majored in biology and minored in chemistry. Then, in my senior year, my professor at Knoxville—he was interested in diversity—got me into a research program at the University of Tennessee, which was not integrated at that time—blacks couldn't attend. But I was allowed to do research on tapeworms, irradiate them and examine their chromosomes, and I was allowed to attend the seminars. I was fascinated by the research, I was just turned on—finally, I realized this is what I'd really like to do.

> ***"One of us from rural America had to make it—and I thought, 'Why not me?'"***

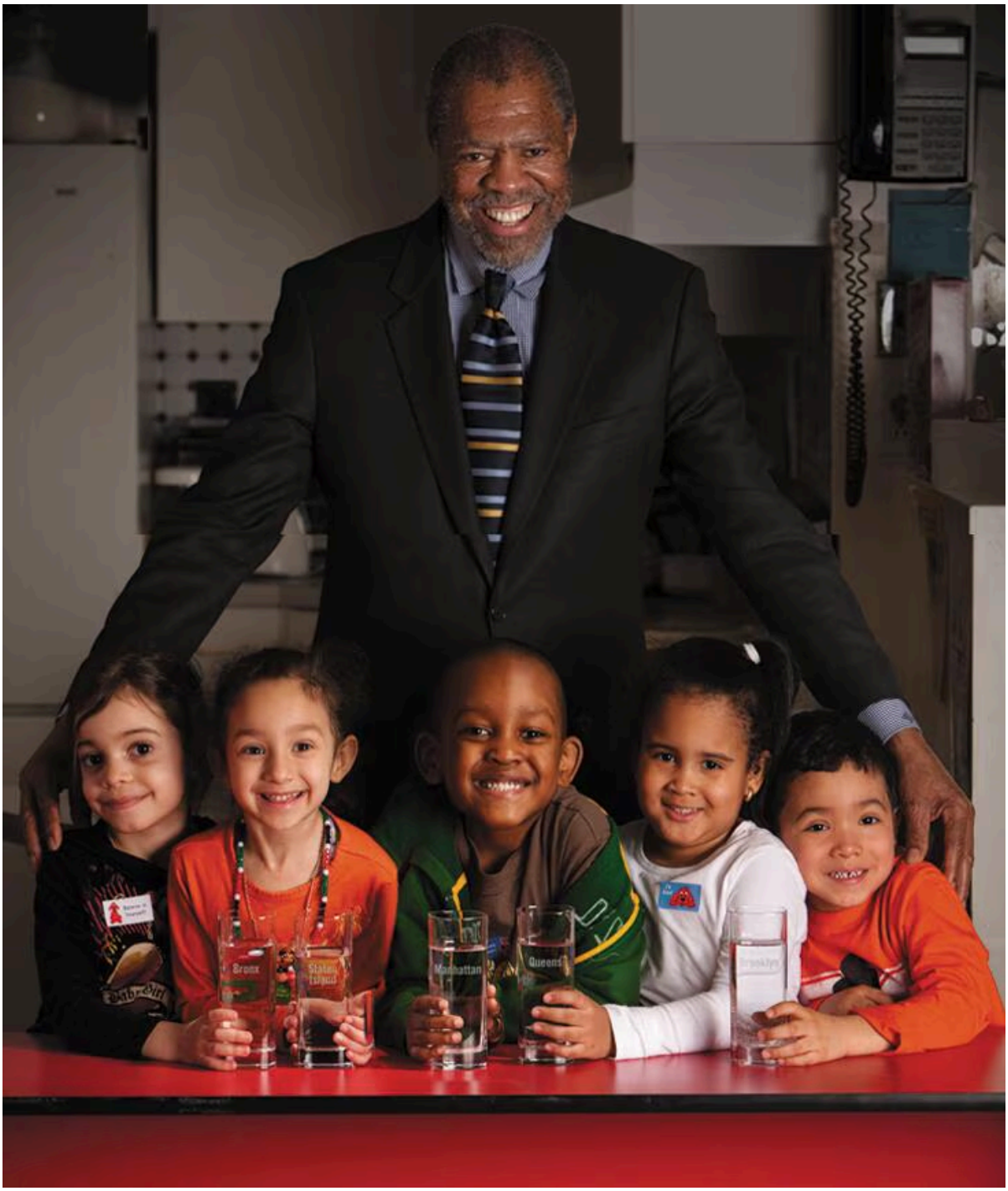
Can you tell me about how you got into cancer research?

After my Ph.D. and my postdoctoral research at Harvard, I realized I wanted to work on animal cells, so I joined Ira Pastan's group at the National Cancer Institute at the National Institutes of Health, where I eventually got my own lab. Together with Ken Yamada, I was working on a protein called fibronectin, which was present on the outside surface of normal cells but not cancer cells. Fibronectin is a glycoprotein—it has carbohydrates (sugars) attached to it. At the time, there was a hypothesis that the carbohydrates were necessary for fibronectin to be exported from the cell, and we decided to test that hypothesis using a drug called tunicamycin that prevented carbohydrate attachment. We showed that carbohydrates weren't necessary for export, but instead they were important for stabilizing the protein's structure. That turned out to be one of the most cited papers for 1978; it was huge.

In 1991, you became Director of the National Institute of Environmental Health Sciences. What were your goals and accomplishments there?

When I interviewed for the position, I told the Director of the NIH, "My first priority would be to make the Institute responsive to the needs of the American people." She immediately offered me the job—and that changed my life. It gave me the opportunity to think big and to address a lot of issues I felt weren't being

dealt with, kind of what I'd been dreaming of. Environmental health research at that time focused on chemical carcinogenesis, and I wanted to expand that focus to social and behavioral issues also, as well as genetics. Over my time there, I engaged communities in identifying areas of concern for our research, such as disproportionate exposure to chemicals in certain neighborhoods. I founded the Environmental Genome Project, which used a novel genomic approach to determine susceptibility to toxins. I also expanded Environmental Health Centers around the country, developing the Breast Cancer and the Environment Research Program and Children's Environmental Health and Disease Prevention Research Centers. Children are really important to me—they are especially susceptible to environmental toxins, and we needed to address that.



Dr. Olden established Children's Environmental Health and Disease Prevention Research Centers.

What is your advice to an undergraduate considering a career in biology?

Most people, I think, will figure out what is the right thing to do, but often it takes a lot of courage to do the right thing. When I accepted the Sackler Prize, I talked about walking to high school and realizing that government was making a lot of decisions that affected rural America without ever bothering to consult rural Americans. In order to change that, one of us from rural America had to make it—and I thought, "Why not

me?” In being awarded the prize, for creating community-based participatory research, it looks like I actually succeeded in what I set out to do: to get the public health decision-makers to pay attention to the needs of the poor.